LAB # 2
To Analyse and Explain the Syntax and Semantics of Application Layer Service and Protocols using Wireshark

Objectives

80 25 110 53

- To analyse and show the contents captured in Wireshark related to HTTP, SMTP, POP and DNS.
- To show the network transactions between clients and DNS server using Wireshark.

Pre-Lab Exercise

Read this experiment in its entirety to become familiar with objectives of this lab. Study in detail and become familiar with the basics of Hyper Text Transfer Protocol (HTTP), Simple Mail Transfer Protocol (SMTP) and Domain Name Server (DNS) provided with this laboratory experiment and in chapter 2 of the reference book. Have Wireshark Packet Sniffer software program installed on your PC and review how to capture/analyse packets. You may record the terms and sections that require more elaboration for reference. The instructor may provide the class some time to reflect upon these before proceeding with the lab.

Introduction to Packet Sniffer

One's understanding of network protocols can often be greatly deepened by "seeing protocols in action" and by "playing around with protocols" – observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences. This can be done in simulated scenarios or in a "real" network environment such as the Internet. In the Wireshark labs you'll be running various network applications in different scenarios using your own computer. You'll observe the network protocols in your computer "in action," interacting and exchanging messages with protocol entities executing elsewhere in the Internet. Thus, you and your computer will be an integral part of these "live" labs. You'll observe, and you'll learn, by doing.

In this first Wireshark lab, you'll get acquainted with Wireshark, and make some simple packet captures and observations.

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures ("sniffs") messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure 5 shows the structure of a packet sniffer. At the right of Figure 5 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure 5 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. Recall from the discussion from the first lab that messages exchanged by higher layer protocols such as HTTP, FTP, TCP, UDP, DNS, or IP all are eventually encapsulated in link-layer frames that are transmitted over physical media such as an Ethernet cable. In Figure 5 , the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

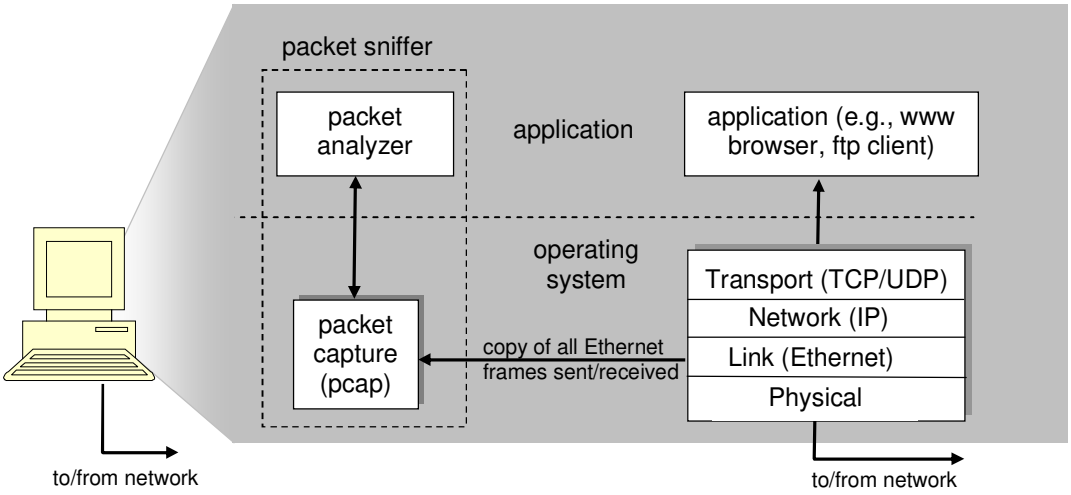

Figure 5: Packet Sniffer Structure

The second component of a packet sniffer is the **packet analyser**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyser must "understand" the structure of all messages exchanged by protocols.

We will be using the Wireshark packet sniffer for these labs, allowing us to display the contents of messages being sent/received from/by protocols at different levels of the protocol stack.

Running Wireshark

When you run the Wireshark program, you'll get a startup screen

Take a look at the upper left hand side of the screen – you'll see an "Interface list". This is the list of network interfaces on your computer. Once you choose an interface, Wireshark will capture all packets on that interface. Start Wireshark on your computer system is there an Ethernet interface?

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yes there is, but its not in active state because my system is connected through wifi (wlp2s0)

Click on Ethernet interface and then start. A screen like the one below will be displayed, showing information about the packets being captured. Once you start packet capture, you can stop it by using the Capture pull down menu and selecting Stop.

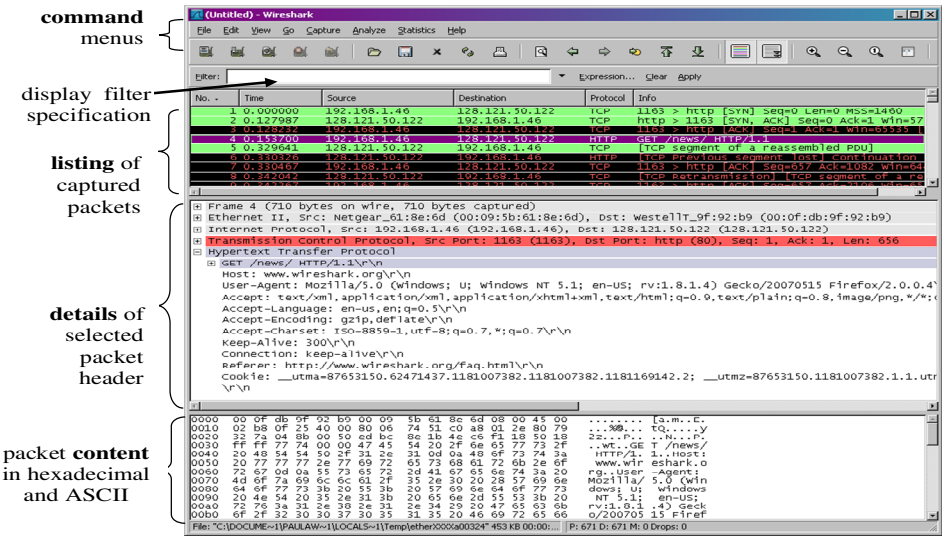

Figure 6: Wireshark GUI, during packet capture and analysis

- The Wireshark interface has five major components:
- The **command menus** are standard pulldown menus located at the top of the window. Of interest to us now are the File and Capture menus. The File menu allows you to save captured packet data or open a file containing previously captured packet data, and exit the Wireshark application. The Capture menu allows you to begin packet capture.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number (assigned by Wireshark; this is *not* a packet number contained in any protocol's header), the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet. The packet listing can be sorted according to any of these categories by clicking on a column name. The protocol type field lists the highest-level protocol that sent or received this packet, i.e., the protocol that is the source or ultimate sink for this packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. (To select a packet in the packet-listing window, place the cursor over the packet's one-line summary in the packet-listing window and click with the left mouse button.). These details include information about the Ethernet frame (assuming the packet was sent/received over an Ethernet interface) and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.
- Towards the top of the Wireshark graphical user interface, is the **packet display filter field,** into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window (and hence the packet-header and packet-contents windows). In the example below, we'll use the packet-display filter field to have Wireshark hide (not display) packets except those that correspond to HTTP messages.

Task 1: Starting with Wireshark

Step 1: Check if your computer is connected to the Internet via a wired Ethernet interface? Do the following
Start up your favorite web browser, which will display your selected homepage.
Start up the Wireshark software.
To begin packet capture, select the *Capture* pull down menu and select *Interfaces*. This will cause the "Wireshark: Capture Interfaces" window to be displayed, as shown in Figure 7.

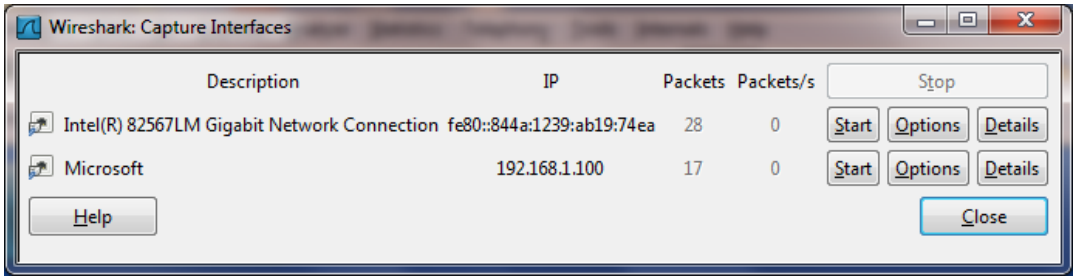

Figure 7: Wireshark Capture Interface Window

Select Ethernet interface.
Once you begin packet capture, a window similar to that shown in Figure 2 will appear. This window shows the packets being captured. By selecting *Capture* pulldown menu and selecting *Stop*, you can stop packet capture.
We will capture some interactive packets and analyse packets in next task.

In-Lab Exercise

Task 1: Capture and Analyse HTTP Trace

Step 1: Let's begin our exploration of HTTP by accessing a website. Do the following:

- Start up your web browser.
- Enter "http" (just the letters, not the quotation marks) in the display-filter-specification window, so that only captured HTTP messages will be displayed later in the packet-listing window.
- Wait a bit more than one minute (we'll see why shortly), and then begin Wireshark packet capture.
- Enter the address to your browserhttps://www.google.com.pk
- As web page appear on browser, go to Wireshark window and stop packet capture.

What do you observe in *packet-listing window*?

Packet listing window shows all the packets capture by pc wifi. It have unique capture number,

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timestamp, source/destination IPs, protocol, and packet info .

Step 2: Explore the packets and answer the following questions

What version of HTTP is your browser and google server running?

_____._____

HTTP/1.1

_____

What languages (if any) does your browser indicate that it can accept to the server?

en-gb , en-us

What is the IP address of your computer?

10.135.40.197

What is the IP address of google server?

128.119.245.12

How many bytes of content are being returned to your browser?

375 bytes

What is the source and destination port number?

src : 80 (static) dest : 34754 -> response reference

How long did it take from when the HTTP GET message was sent until the HTTP OK reply was received?

3.41925 - 3.01163 = 0.40762

What is date and time of server in HTTP received packets?

Wed, 18 Feb 2026 10:36:09 GMT

Task 2: Analyse SMTP Trace

Step1: Download the following file, and open it up in Wireshark:

http://asecuritysite.com/log/smtp.zip

Step 2: Explore the packets and answer the following questions

The IP address and port used by the host which is sending the Email

192.168.0.12:1713

The IP address and the port used by the SMTP server

192.168.0.13:25

Who is sending the Email?

martin.tor@4salet.com

Who is receiving the Email?

bert.manly@five8nine.com

When was the Email sent?

Date: Mon, 11 Mar 2013 22:10:34

What was the message, and what was the subject of the Email?

subject : T2theS4gSSB0aGluay (encrypted) message :

With SMTP, which character sequence is used to end the message?

<CR><LF>.<CR><LF>

Task 3: Analyse POP3 trace

Step 1: Download the following file, and open it up in Wireshark

http://asecuritysite.com/log/pop3.zip

Step 2: Explore the packets and answer the following questions

The IP address and the port used by the host which is sending the email

192.168.0.4:26386

The IP address and the port used by the POP-3 server

212.227.15.166:110

Whose mail box is being accessed?

digitalinvestigator@networksims.com

How many email messages are in the Inbox?

3 check response of the info of stat

Which command does POP-3 use to get a specific message?

RETR <message no>

For Message 1, who sent the message and what is the subject and outline the content of the message?

From: 1&1 Internet Ltd. <support@1and1.co.uk>
Subject: A message from 1&1 Internet
To: digitalinvestigator@networksims.com
outline: We'd like to take this opportunity to tell you about a feature . . .

Which command does POP-3 use to get a specific message?

RETR <message no>

i am using linux, so i used the relevant commands closest to that command prompt's commands

DNS

The Domain Name System (DNS) translates hostnames to IP addresses, fulfilling a critical role in the Internet infrastructure. In this lab, we'll take a closer look at the client side of DNS. Client's role in the DNS is relatively simple – a client sends a *query* to its local DNS server, and receives a *response* back.

In this lab we will get DNS record and analyse it. Also trace the packets between your computer and DNS server.

fall back hierarchy:Authoritative dns (cuilahore.edu.com) -> Top level domain TLD server(.edu.com) -> (Recursive Resolver (ISP) (caches it) -> Your OS Cache (caches it) -> Browser

Task 1: Get DNS record

To send query any specified DNS server for a DNS record *nslookup* tool can be used. Your internet browsers already programmed to get DNS records. *nslookup* tool allows user to analyse the DNS records. To run it in Windows, open the Command Prompt and run *nslookup* on the command line.

Step 1: Open Command Prompt and run following command and write down your observations

*nslookup*www.google.com

muhammad-ahmad@latitude-7490:~$ nslookup google.com

1. local dns cache resolve the query	Server: 127.0.0.53 Address: 127.0.0.53#53
2. received non-authoritative answer from google.com	Non-authoritative answer: Name: google.com Address: 142.250.187.46
3. it shows A records and AAAA records for google.com	Name: google.com Address: 2a00:1450:4018:80c::200e

Step 2: Run following command and write down your observations

nslookup –type=NS www.google.com

muhammad-ahmad@latitude-7490:~$ nslookup -type=NS google.com

	Server: 127.0.0.53 Address: 127.0.0.53#53
1. local dns cache resolve the query	Non-authoritative answer:
2. received both non-authoritative and authoritative answer from google.com	google.com nameserver = ns4.google.com. google.com nameserver = ns1.google.com. google.com nameserver = ns3.google.com. google.com nameserver = ns2.google.com.
3. this also shows all name servers (NS) (ns1,ns2,ns3,ns4) availabe for google.com	Authoritative answers can be found from:
4. it also shows A records and AAAA records for each name server	ns2.google.com internet address = 216.239.34.10 ns2.google.com has AAAA address 2001:4860:4802:34::a ns4.google.com internet address = 216.239.38.10 ns4.google.com has AAAA address 2001:4860:4802:38::a ns1.google.com internet address = 216.239.32.10 ns1.google.com has AAAA address 2001:4860:4802:32::a ns3.google.com internet address = 216.239.36.10 ns3.google.com has AAAA address 2001:4860:4802:36::a

Step 3: Run following command and write down your observations

You can indicate that you want to the query sent to any DNS server rather than to the default DNS server. Like in the following example we send query to google public DNS server. Thus, the query and reply transaction takes place directly between our querying host and 8.8.8.8.

*nslookup*www.hotmail.com8.8.8.8

muhammad-ahmad@latitude-7490:~$ nslookup hotmail.com 8.8.8.8

1. query is resolved by our mentioned DNS server	Server: 8.8.8.8 Address: 8.8.8.8#53
2. received non-authoritative answer from mentioned DNS server	Non-authoritative answer: Name: hotmail.com
3. it only show A records	Address: 204.79.197.212

Task 2: Get your system address

ipconfig is among the most useful little utilities in your host, especially for debugging network issues. ipconfig can be used to show your current IP information, including your address, DNS server addresses, adapter type and so on.

Step 1: Run the following command in Command Prompt and write down your observations

ipconfig /all

linux commands

ip addr

ip addr show

resolvectl status : dns info

there are 3 different interfaces

1. lo (loopback) -> localhost
2. enp0s31f6 -> ethernet (not connected)
3. wlp2s0 -> wireless (connected)
   - i- state is up
   - ii- ipv6 : fe80::b272:bf75:67f3:4cc9/64
   - iii- ipv4 : 10.135.88.78/22

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defa
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
      valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
      valid_lft forever preferred_lft forever

2: enp0s31f6: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state D
    link/ether e4:b9:7a:6c:54:4e brd ff:ff:ff:ff:ff:ff

3: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP gro
    link/ether 98:3b:8f:04:b2:5d brd ff:ff:ff:ff:ff:ff
    inet 10.135.88.78/22 brd 10.135.91.255 scope global dynamic noprefixroute wlp2s0
      valid_lft 3490sec preferred_lft 3490sec
    inet6 fe80::b272:bf75:67f3:4cc9/64 scope link noprefixroute
      valid_lft forever preferred_lft forever
```

Step 2: Run the following command in Command Prompt and write down your observations

ipconfig /displaydns

linux command : resolvectl statistics

1- transaction : currently no dns query (transaction) is pending overall 5277 queries resolved since the system booted

2- total of 35 dns lookup are cached with hit ratio of 858/4199

3- total 13 queries faild due to net issues, 3 are unvalid

```
muhammad-ahmad@latitude-7490:~$ sudo resolvectl statistics
[sudo] password for muhammad-ahmad:
Transactions
                Current Transactions:    0
                  Total Transactions: 5277
Cache
                  Current Cache Size:   35
                        Cache Hits:  858
                      Cache Misses: 4199
Failure Transactions
                    Total Timeouts:   13
        Total Timeouts (Stale Data Served):    0
                Total Failure Responses:    3
Total Failure Responses (Stale Data Served):    0
DNSSEC Verdicts
                          Secure:    0
                        Insecure:    0
                           Bogus:    0
                   Indeterminate:    0
```

Step 2: Run the following command in Command Prompt and write down your observations
ipconfig /flushdns

linux command : resolvectl flush-caches

after clearing cache

Current Cache Size: 3
Cache Hits: 878
Cache Misses: 4247

it clear all the cached dns , we clearly see that the cache size droped from 35 to 3

Task 3: Tracing DNS with Wireshark

Step 1: Do the following

Use *ipconfig* to empty the DNS cache in your host.
Open your browser and empty your browser cache.
Open Wireshark and enter "ip.addr == your_IP_address" into the filter, where you obtain your_IP_address with ipconfig. This filter removes all packets that neither originate nor are destined to your host.
Start packet capture in Wireshark.
With your browser, visit the Web page: http://www.ietf.org
Stop packet capture.

Step 2: Analyse the DNS packets on Wireshark and answer the following questions

Locate the DNS query and response messages. Are then sent over UDP or TCP?

udp

What is the destination port for the DNS query message? What is the source port of DNS response message?

1:92.168.0.1:53 port : 53

reslovectl status

To what IP address is the DNS query message sent? Use ipconfig to determine the IP address of your local DNS server. Are these two IP addresses the same?

192.168.0.1 same

Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"?

Type: A (1) (Host Address) : ip4 / ans : no

Examine the DNS response message. How many "answers" are provided? What do each of these answers contain?

ans = 1 query send to ipv4 internet ip addr
ans : rr1---sn-npoldne7.googlevideo.com: type A, class IN, addr 142.251.91.134

Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message?

Yes, the destination IP in the SYN packet corresponds to the IP address provided in the DNS response.

DNS Response → 142.251.91.134
TCP SYN → Sent to 142.251.91.134

Rubric for Lab Assessment

The student performance for the assigned task during the lab session was:			
Excellent	The student completed assigned tasks without any help from the instructor and showed the results appropriately.	4	
Good	The student completed assigned tasks with minimal help from the instructor and showed the results appropriately.	3	
Average	The student could not complete all assigned tasks and showed partial results.	2	
Worst	The student did not complete assigned tasks.	1	

Instructor Signature: _____ Date: _____